

TORILO ACADEMY
Data Analysis Program — Charis Cohort

CAPSTONE PROJECT — PART B

Consolidated Analysis Report

Covering the SQL, Power BI, and Excel Module Projects

Prepared by: Olaseni Kamorudeen Lawal

Cohort: Charis Cohort

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1. Executive Summary

This report consolidates the three module projects completed during the Data Analysis Program at Torilo Academy: a SQL workforce analytics project, a Power BI dashboard project, and an Excel data project. Each project targeted a different stage of the analytics workflow — from cleaning and querying relational data in MySQL, to designing an interactive business intelligence dashboard in Power BI, to building formulas, PivotTables, and charts in Excel. Together, they demonstrate a working command of data cleaning, transformation, querying, visualization, and business storytelling.

Project	Tool	Dataset	Headline Output
SQL	MySQL Workbench	Orion Data Systems workforce (107 employees, 27 departments, 25 countries)	8 answered business questions on headcount, salary, and country presence
Power BI	Power BI Desktop	1,074 global unicorn companies	5-visual dashboard; \$3.711T total market cap
Excel	Microsoft Excel	Trendora shopping data, 3,900 customers	4-chart dashboard; \$233,081 total purchase value

2. Project 1 — SQL: Orion Data Systems Workforce Analytics

2.1 Background & Objective

Orion Data Systems is a multinational consulting and technology firm headquartered in San Francisco, with offices across Europe, Asia, and the Americas. As the assigned Junior Data Analyst, the objective was to answer eight business questions for the HR & Strategy team using SQL, covering headcount distribution, salary structure, and country-level workforce presence.

2.2 Data & Tools

Tool: MySQL Workbench (schema: capstone)

Tables: employees (107 rows), departments (27 rows), jobs (19 rows), countries (25 rows)

Techniques used: JOINS, GROUP BY, aggregate functions (COUNT, AVG, SUM), CASE statements, subqueries, Common Table Expressions (CTEs), LEFT JOIN for gap analysis

2.3 Data Cleaning Steps

- Removed a UTF-8 byte-order-mark (BOM) character that had corrupted the country_id column header on import.
- Standardized primary key column names across all four tables so joins would resolve correctly.
- Converted hire_date from a text field in M/D/YYYY format to a proper DATE type, using STR_TO_DATE, then verified zero NULL conversions before dropping the original text column.
- Verified table structures with DESCRIBE after each change to confirm data types before analysis.

2.4 Key Findings

Workforce Distribution — Shipping is by far the largest department with 45 employees, followed by Sales with 34. Together these two departments account for over 73% of the 107-person workforce, while Administration, Human Resources, and Public Relations each operate with a single employee.

Department	Employee Count
Shipping	45

Department	Employee Count
Sales	34
Purchasing	6
Finance	6
IT	5
Executive	3

Salary Comparison — The Executive department has by far the highest average salary at 19,333, while Shipping has the lowest at 3,476, consistent with it being the highest-headcount, most operational department.

Department	Average Salary
Executive	19,333.33
Accounting	10,150.00
Public Relations	10,000.00
Marketing	9,500.00
Sales	8,955.88
Shipping	3,475.56

Salary Bands — Employees classified as Low (<5,000), Medium (5,000–10,000), or High (>10,000): 49 Low, 43 Medium, 15 High.

Country-Level Analysis — Orion operates across 25 countries. Switzerland hosts the widest departmental footprint (5 departments), followed by the United States, Japan, Italy, and Australia (4 each).

High Earners — The company-wide average salary is 6,461.68; 51 of the 107 employees (about 48%) earn above this average.

Job Role Analysis — Four job titles average above 12,000: President (24,000), Administration Vice President (17,000), Marketing Manager (13,000), and Sales Manager (12,200).

Salary Growth Trend — Switzerland carries the highest total salary cost (75,100), followed by Italy (52,100) and Israel (45,200).

Workforce Gaps — Every job role defined in the jobs table currently has at least one employee assigned; no vacant job codes exist.

2.5 Business Recommendations

- Shipping's disproportionate headcount and low average pay suggest reviewing whether roles here are undervalued relative to responsibility.
- Switzerland's combination of high department count and highest total salary cost marks it as Orion's key strategic hub for retention and succession planning.
- With 48% of staff earning above the company average, HR could use the salary-band segmentation as a starting point for a formal compensation-banding review.

3. Project 2 — Power BI: The Billion Dollar Club (Global Unicorn Market)

3.1 Background & Objective

Unicorns — startups valued at \$1 billion or more before going public — are a widely used barometer of where venture capital and innovation are concentrating globally. The objective was to build a Power BI dashboard letting a stakeholder

quickly see the size of the unicorn market, which industries and countries dominate it, how fast new unicorns emerge, and how funding relates to valuation at the company level.

3.2 Data & Tools

Tool: Power BI Desktop

Source data: Unicorn_Companies dataset — 1,074 companies, 10 fields

Techniques used: Power Query (M code) for currency-string cleaning, a dedicated _Measures table for DAX, KPI cards, line chart, bar chart, scatter plot, filled map, and slicers

3.3 Data Preparation (Power Query)

- Stripped the "\$" symbol and thousands separators from the Valuation and Funding columns.
- Parsed the trailing unit letter (B, M, K) and multiplied the numeric portion accordingly, using M code, to produce true numeric columns.
- Derived a "Years to Scale" column as Date Joined (year) minus Year Founded.
- Confirmed zero missing Valuation values across all 1,074 rows; 12 rows had missing Funding data, left blank rather than imputed.

3.4 Dashboard Design

- Four KPI cards: Average Valuation, Average Years to Scale, Total Market Cap, Total Companies.
- A "New Unicorns Per Year" line chart.
- A "Total Market Cap by Industry" horizontal bar chart.
- A "Funding and Valuation by Company" scatter plot on log-scaled axes.
- A "Total Market Cap by Country" filled/bubble map, plus Industry and Continent slicers.



Figure 1: The Billion Dollar Club — Global Unicorn Market dashboard (Power BI)

3.5 Key Findings

Market Size — A combined market cap of \$3.711 trillion, an average valuation of \$3.46 billion, and 1,074 total unicorn companies (rounded to "1K" on the dashboard card). Companies take an average of 7 years from founding to reach unicorn status.

Growth Trend — New unicorn creation rose steadily through 2018–2020 (103 to 108 per year), then spiked sharply to 520 in 2021 before falling back to 116 in 2022.

Year Joined	New Unicorns
2018	103
2019	104
2020	108
2021	520
2022	116

Industry Concentration — Fintech leads by total valuation (\$882bn) and by company count (224), followed by Internet Software & Services (\$595bn, 205 companies), E-commerce & Direct-to-Consumer (\$426bn), and Artificial Intelligence (\$361bn).

Geographic Concentration — The United States dominates (\$1.933 trillion, 562 companies), followed by China (\$696bn, 173 companies) and India (\$196bn, 65 companies). North America and Asia together account for the vast majority of global unicorn value.

Top Companies — The five highest-valued unicorns are Bytedance (\$180bn), SpaceX (\$100bn), SHEIN (\$100bn), Stripe (\$95bn), and Klarna (\$46bn), spread across China, the US, and Sweden.

3.6 Business Recommendations

- Fintech is the most competitive but most "proven" category; Artificial Intelligence (fewer companies, high valuation concentration) may offer more room for a new entrant to stand out.
- The 2021 spike is likely an anomaly tied to the broader venture funding boom that year, so growth projections should use a multi-year average rather than the 2021 figure alone.
- A go-to-market or investment strategy focused on the US and China captures most of the addressable opportunity, with India and the UK as meaningful secondary markets.

4. Project 3 — Excel: Trendora Sales & Customer Insights

4.1 Background & Objective

Trendora is a retail business whose transaction-level shopping data — covering 3,900 customers — was analyzed in Excel to understand purchasing patterns, customer demographics, payment preferences, and seasonal demand. The objective was to build a single-view Excel dashboard that a sales or merchandising team could use to spot which products, seasons, age groups, and payment methods drive the business.

4.2 Data & Tools

Tool: Microsoft Excel

Source data: Trendora shopping_data sheet — 3,900 rows, 17 fields (Customer ID, Age Group, Gender, Item Purchased, Category, Purchase Amount, Location, Season, Review Rating, Subscription Status, Shipping Type, Promo Code Used, Payment Method, Discount value, and more)

Techniques used: PivotTables/summary tables per metric (top items, payment method, season, age group), a dedicated Measures sheet, and a KPI-card-and-chart dashboard layout combining a bar chart, pie chart, line chart, and column chart

4.3 Dashboard Design

- Three KPI cards at the top: Total Purchase Amount, Average Review Rating, and Total Number of Customers.
- A "Top 5 Items Purchased" horizontal bar chart.
- A "Payment Method Breakdown" pie chart.
- A "Purchase Amount by Season" line chart.
- A "Purchase Distribution by Age Group" column chart.
- Payment Method, Season, and Gender slicers on the right, letting the dashboard filter interactively.

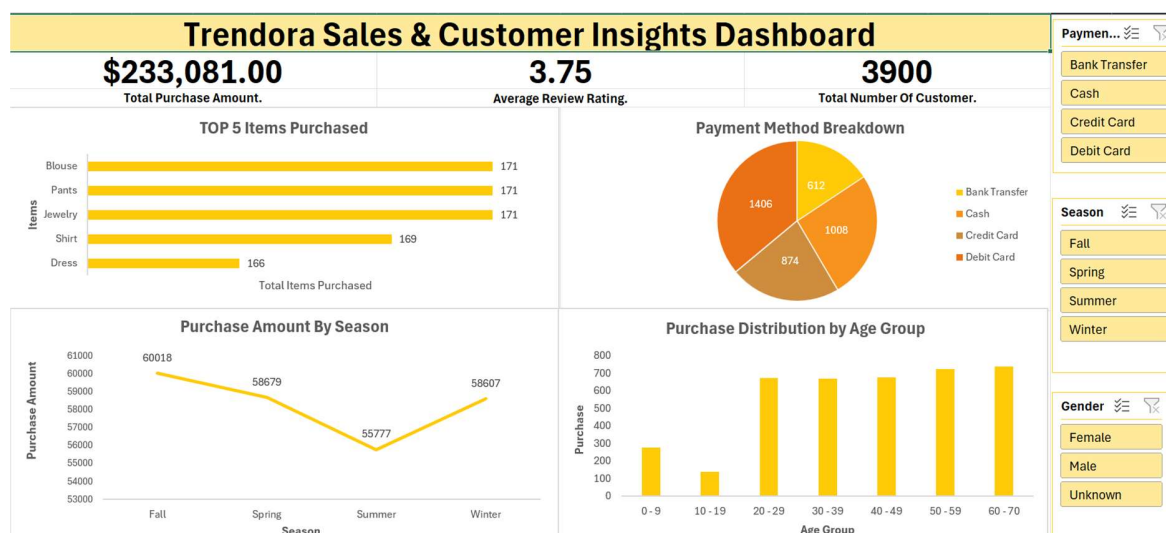


Figure 2: Trendora Sales & Customer Insights dashboard (Excel)

4.4 Key Findings

Overall Volume — The dashboard's three KPI cards are confirmed directly against the underlying data: \$233,081 total purchase amount, a 3.75 average review rating, and 3,900 total customers.

Top Products — Blouse, Pants, and Jewelry are tied as the most purchased items (171 units each), with Shirt (169) and Dress (166) close behind — indicating a fairly even spread across Trendora's top five product lines rather than one dominant item.

Item	Units Purchased
Blouse	171
Pants	171
Jewelry	171
Shirt	169
Dress	166

Payment Preferences — Debit Card is the most-used payment method (1,406 transactions, 36% of all purchases), followed by Cash (1,008), Credit Card (874), and Bank Transfer (612), showing customers lean toward direct, non-credit payment options.

Payment Method	Transaction Count
Debit Card	1,406
Cash	1,008
Credit Card	874
Bank Transfer	612

Seasonality — Fall generates the highest purchase volume (\$60,018), followed by Spring (\$58,679) and Winter (\$58,607), while Summer is the lowest (\$55,777) — a relatively narrow \$4,241 spread between the strongest and weakest season, suggesting demand is fairly stable year-round with a modest fall/winter lift.

Season	Purchase Amount (\$)
Fall	60,018
Spring	58,679
Winter	58,607
Summer	55,777

Age Group Distribution — Purchase activity is lowest among the youngest customers (10–19: 138 purchases; 0–9: 276) and climbs steadily from age 20 onward, peaking in the 50–59 (726) and 60–70 (739) brackets — Trendora's customer base skews toward older, likely higher-disposable-income shoppers rather than teens or young children.

Age Group	Purchase Count
0 - 9	276
10 - 19	138
20 - 29	674
30 - 39	669
40 - 49	678
50 - 59	726
60 - 70	739

Category Mix — Clothing is the largest category (1,737 purchases), followed by Accessories (1,240), Footwear (599), and Outerwear (324). Interestingly, average spend per purchase is fairly flat across categories (roughly \$57–\$60), so Clothing's lead is driven by purchase frequency, not higher-value items.

Customer Behavior — Only 1,053 of 3,900 customers (about 27%) are on a subscription, and just 1,024 (about 26%) used a promo code, together representing \$16,790 in total discounts given — indicating meaningful room to grow both subscription enrollment and promo engagement.

As a data-quality check ahead of building the dashboard, promo-code and discount records were independently spot-checked in MySQL by querying the same dataset for customers who used a promo code and received a discount greater than zero, confirming the figures above were consistent across tools.

The screenshot shows a MySQL IDE interface. The left sidebar displays the 'SCHEMAS' panel with a tree view containing 'charis', 'new_schema', 'sample_data', and 'sys'. The main window shows a SQL query editor with the following text:

```
-- . Retrieve all customers who used a promo code and got a discount greater than 0
SELECT *
FROM trendora
WHERE PromoCodeUsed = 'Yes' AND DiscountValue > 0;
```

Below the query editor is the 'Result Grid' tab, which displays a table of results. The table has 12 columns: CustomerID, Age Group, Gender, Item Purchased, Category, PurchaseAmount, Location, Size, Color, Season, ReviewRating, and Subco. The results show 5 rows of data:

CustomerID	Age Group	Gender	Item Purchased	Category	PurchaseAmount	Location	Size	Color	Season	ReviewRating	Subco
1	50 - 59	Female	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes
3	50 - 59	Female	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes
5	40 - 49	Female	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes
8	20 - 29	Female	Shorts	Clothing	34	Louisiana	L	Charcoal	Winter	3.2	Yes
16	60 - 70	Female	Skirt	Clothing	81	Rhode Island	M	Teal	Winter	2.8	Yes

At the bottom of the IDE, there is a tab bar showing multiple 'trendora' tabs, with 'trendora 31' currently selected.

Figure 3: MySQL spot-check confirming promo-code and discount records ahead of the Excel dashboard build

4.5 Business Recommendations

- With Debit Card and Cash together covering 62% of transactions, Trendora should ensure checkout and delivery experiences are frictionless for non-credit payment paths rather than over-indexing promotions toward credit card users.
- Given Summer is the weakest season, a targeted mid-year promotion or product refresh could help close the roughly \$4,200 gap versus Fall.
- Since the 50+ age bracket drives the most purchases, marketing and product assortment could lean further into that segment, while a lighter-touch, value-oriented campaign might help lift the currently small 10–19 segment.
- With subscription and promo-code usage each under 30%, targeted campaigns to convert one-time buyers into subscribers represent a clear, low-risk growth lever.

5. Cross-Project Reflections & Skills Demonstrated

Across the three projects, a consistent analytical workflow is visible: understand the business question, clean and validate the data, apply the right tool-specific technique to answer it, and translate the result into a decision-relevant insight.

Skill Area	SQL Project	Power BI Project	Excel Project
Data Cleaning	BOM removal, date type conversion, column renaming	Currency string parsing via Power Query/M	Structuring 3,900 raw transaction rows into summary tables
Core Technique	Joins, CTEs, subqueries, CASE	DAX measures, filled maps, KPI cards	PivotTable-style summaries, KPI cards, multi-chart dashboard
Output	8 answered business questions	Interactive 5-visual dashboard	4-chart dashboard with slicers

6. Conclusion

The three module projects, taken together, demonstrate an end-to-end data analysis skill set: relational database querying and cleaning in MySQL, interactive dashboard design in Power BI, and spreadsheet-based analysis and visualization in Excel. The SQL project's findings point HR & Strategy leadership toward Shipping's compensation structure and Switzerland's role as a strategic hub. The Power BI project shows a \$3.711 trillion global unicorn market concentrated in Fintech and led by the US and China. The Excel project shows Trendora's \$233,081 in tracked purchases skewing toward Debit Card payments, Fall seasonality, and older customers, with clear room to grow subscriptions and promo engagement. Across all three, the consistent thread is a disciplined workflow of cleaning data first, applying the right tool-specific technique second, and translating results into a concrete business recommendation third — the core skill set this capstone was designed to demonstrate.